AME 34351 - Check-In 3 - Group 5

A. Team Members and Roles

Michael Doyle - Visualization and Modeling Lead
John Carlson - Deployment/Communication Lead
Peter Dacunto - Data Processing Lead
Sophia Bracken - Problem/Domain Lead

B. Link to Google Colab Notebook

https://drive.google.com/file/d/15Ql8WpGE4_A0qux9pJI6yTcxkGlKtCTS/view?usp=sharing

1. Final Problem Statement

The team's completed machine learning model app will be used to model the heating load and cooling load of residential buildings using NREL ResStock, a simulated data set of residential buildings in the United States. The app will serve as a tool for architects and engineers to predict the energy consumption of residential buildings they are designing, with the goal of creating the most energy efficient homes and apartments. Users will be able to input seven parameters, including the state, geometric floor area, temperature setpoint, foundation type, fuel type, number of stories, and period of construction in order to see the predicted heating load, cooling load, and annual cost. The users will be able to use the outputs of the machine learning model in order to make design decisions for the residential building they are planning.

2. Final Data Description - NREL ResStock 2024.2

1. Size of Data:
- Rows -> avg: 194,243.8
    - min: 18,785.0 (WY)
    - max: 975,698.0 (CA)
- Cols -> avg: 381.0
    - min: 381.0
    - max: 381.0

2. Data Features
- Found in the "metadata_and_annual_results" folder
- Hundreds of columns for each dwelling unit, covering:
    - Geometry & envelope
        - Considered: floor area, foundation type, number of stories
        - Ignored: wall/roof/window construction, insulation levels, infiltration, floor, wall, duct, and door area calculations
    - HVAC, water heating and appliances

        - Considered: fuel type
        - Ignored: equipment type and efficiency, duct location/sizing, water heater type (and location, which can vary — attic, mechanical room, crawlspace, garage, basement, living area, outside,
    - Occupant/household characteristics
        - Considered: temperature setpoint properties, construction period
        - Ignored: occupant information, household income, area median income, federal poverty level, tenure status
    - Geography (all ignored; state, county, PUMA, climate zone data)
    - Weights (sample weight column representing how many real-world dwelling units each model represents)

3. Labels and Outputs
- State-by-State Annual Aggregations
    - Considered Outputs: Heating/Cooling Load
        - From the Heating/Cooling Load predictions, further extrapolation/predictions are able to be made for annual energy cost as well as representations of state-by-state average consumption vs individual user input consumption

4. Train/Validation/Test Split
- 70% of data will be used to train the model and help set weights & biases
- 15% of data will be used for validation to set hyperparameters and check for overfitting
- 15% of data will be used to test the finalized model

5. Summary + Locations of Target Parameters for App Development
- In.geometry_floor_area_bin - aggregated composition of floor areas, sorted into various wide bins
- in.heating/cooling_setpoint - idealized temperature that should be set for optimal efficiency in a home
- in.heating/cooling_fuel - marker of the fuel type for a given home
- in.vintage - native ResStock categorization of home construction into various time period bins
- in.geometry_foundation_type - marker of the foundation type for a given home
- in.geometry_stories - number of floors in a building

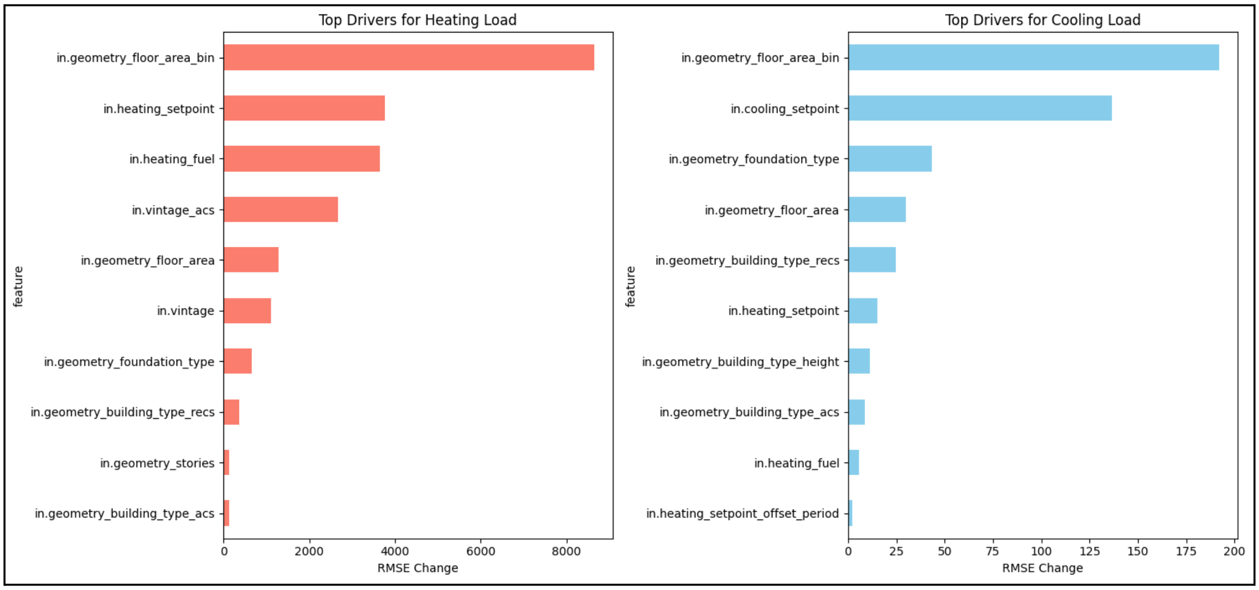
6. The team does not need to perform any additional preprocessing steps on the NREL ResStock 2024.2 dataset.

3. Working Baseline Model

1. Code for the baseline model can be found in the team's Colab notebook.
The core functionality of the code is built on both Ridge Regression and Random Forest, with the bulk of the modeling that will be used for our app built on Random Forest for its improved accuracy. In plot 2a, Random Forest was used to identify the top 10 largest

movers of heating/cooling load, based on each respective attribute's individual impact on RMSE change.

2a. Quantitative Metric: Top Predictors of Heating/Cooling Load

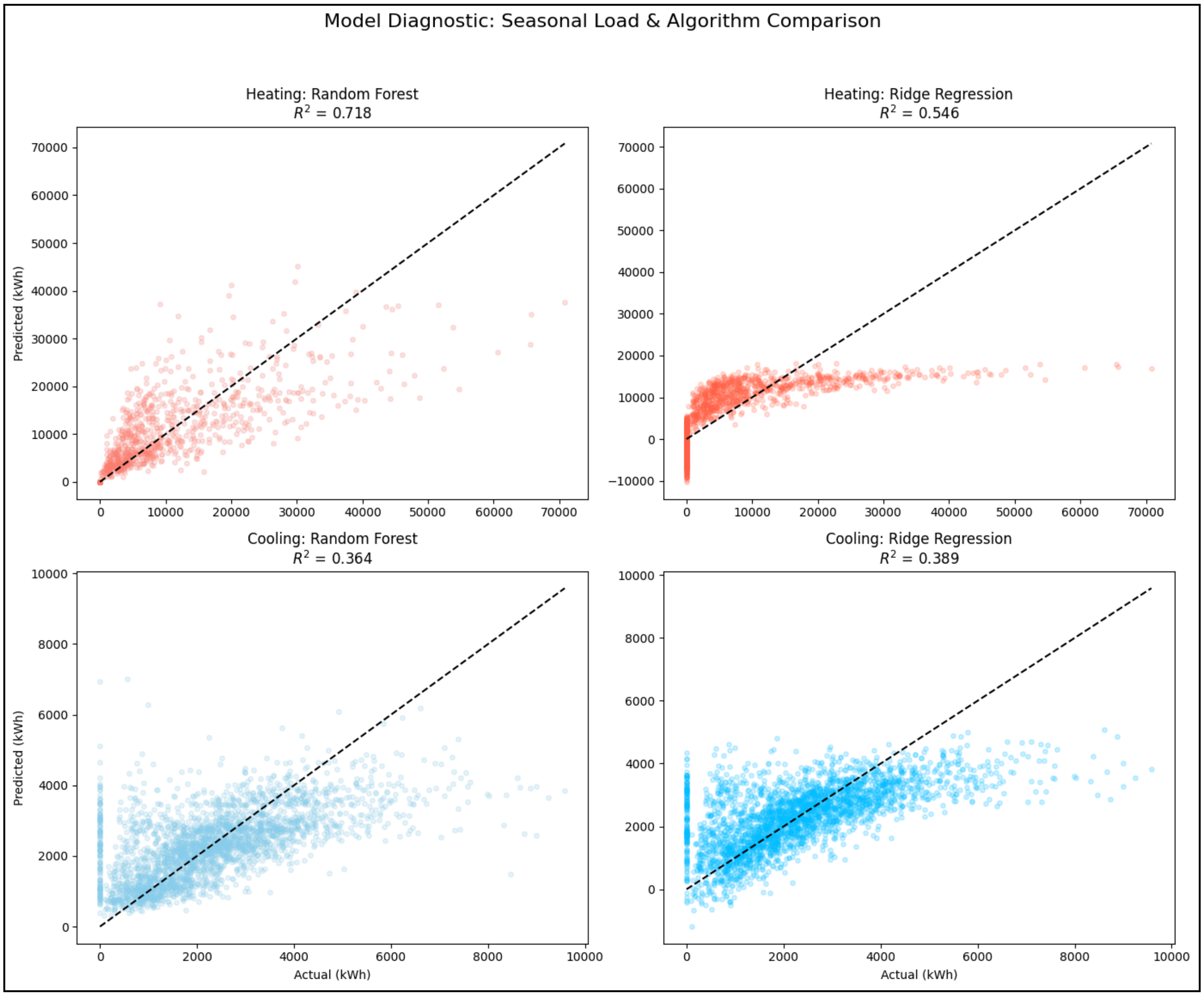


2b. Interpretation of Results

Based on the RMSE change in the prediction model from each input characteristic, these two bar graphs depict the top 10 largest drivers of heating and cooling load. As expected, the total floor area and setpoint are the two most prominent predictors, but other meaningful trends are present as well. For instance, heating fuel has a much more outsized impact on heating load as compared to cooling, indicating that much more of the cooling energy consumption is electric. In addition, the input parameter 'vintage' (a metric of the era a building was constructed in) has a notable impact on heating load as compared to cooling load, indicating that the age of a house has a closer correlation with the heating efficiency.

3a. Quantitative Metric #2: Ridge Regression vs. Random Forest Comparison

In this chart, four different plots depict the accuracy (in the form of predicted vs. actual scatter plots) of ridge and regression for heating and cooling load. While both models struggled generally with predicting cooling load, random forest had a notably better R^2 on both sets, especially heating load.



4. Planned Improved Model over Current Implementation

1. Model Name/Type: K-Nearest Neighbors (KNN) for Cooling Load Prediction
2. Model Description

KNN is a non-parametric, instance-based learning algorithm that works on the principle of similarity. Instead of learning an explicit global function (like Ridge Regression) or building a tree structure (like Random Forest), KNN predicts the energy load of a building by looking at the 'k' most similar buildings in the training set.

3. Why/How It Could Help the Cooling Model

- Cooling loads can be highly sensitive to specific combinations of local factors, like a specific floor area range combined with a specific window-to-wall ratio and climate factors. KNN can effectively capture these local relationships, which global models might miss.
- KNN captures complex interactions between features without requiring any explicit engineering interaction terms (example: floor_area * insulation_level).

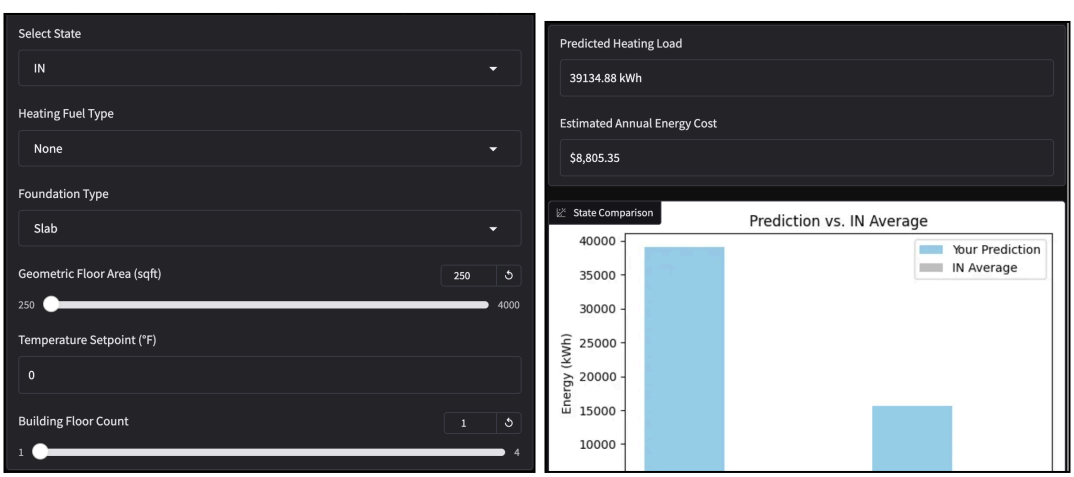
- The model provides a different (simpler and more effective) mathematical approach than the existing Ridge and Random Forest model.

4. Considerations for Proposed Model
   - KNN is extremely sensitive to data scale. The team will need to use Scikit Learn's StandardScaler within a pipeline to ensure that features with large numerical ranges don't unfairly dominate other categorical features.
     - Similar to regularization processes
   - In order for the model to perform its proper functions, the team will need to choose how to measure similarities between data points and choose the correct number of neighbors to consider in each calculation.
     - Number of neighbors = k

5. Final Artifact Description

Who uses it, what inputs it takes, what it displays, and what technology you will use. Include a sketch or wireframe of the interface.

The final artifact for this project will consist of a Gradio-implemented web application built upon the Google Colab Notebook, with six parameter fields for customizing an individual model for a user. The application will consist of drop-down menus for state selection, fuel type, and foundation type. A user will be able to enter numerical text inputs for geometric floor area, temperature setpoint, and period construction. Finally, a user will be able to set their building floor count with a slider. Combining all of these input parameters, the Gradio app will build a model for these particular parameters and predict the associated heating/cooling load as well as predicted annual energy cost. Finally, the Gradio app will plot the user's predicted heating/cooling load versus their input state's average heating/cooling load to provide insights into comparable energy expenditures. The image below is a sample representation of what the user may input and receive in turn.



6. Validation Plan for Final Results

The group will make sure the final results are trustworthy by making sure that the models predictions remain accurate for unseen data and are not influenced by overfitting or data leakage that can occur during training. Since the model has been designed to predict residential building heating and cooling loads in the United States on a state-by-state basis, it is important that the results are able to generalize well to new buildings rather than only performing well on the training data.

These results can be ensured by dividing the dataset into three different sections: 70% training data, 15% validation data, and 15% test data. The training set is used to train the model, the validation set is used to tune the hyperparameters and monitor performance, and the test set will only be used for the final evaluation on unseen data. Both Random Forest and Ridge regression were selected for their robustness against outliers and regularization abilities. Ridge regression is generally useful when features are closely related because it helps prevent the model from relying too much on a single feature, and Random Forest is particularly useful by being insulated from noise, instead relying on "the wisdom of the crowd".

Because ridge regression is sensitive to the scale of the data a standard scaler will be used to normalize the numerical features before training, and this makes sure that the features with larger values, such as the area, do not have more influence on the model than smaller scale features. The model's performance is evaluated using Mean Absolute Error and Root Mean Squared Error. MAE measures the average prediction error and RMSE puts a large emphasis on large errors. The model will check for overfitting by comparing the validation data and training data performance.

To prevent data leakage, steps such as feature scaling will only be fit on the training data before it is applied to the validation and testing datasets. Also, the test dataset will remain separate during model development to make sure that the final performance results will provide an unbiased and trustworthy evaluation of the model.

7. Work Plan for Remainder of Project

The members of the team will work according to the following plan:

- Sophia Bracken (Problem/Domain Lead) - Work on final report and help to debug Python code
- John Carlson (Deployment/Communication Lead) - Design and create app interface, edit written submissions for effective communication
- Peter Dacunto (Data Processing Lead) - Structure and format dataset for final model and report
- Michael Doyle (Visualization/Modeling Lead) - Convert Colab notebook to working app infrastructure, ensure effective implementation of planned model